

SUPHARADA JUNDOK (李沛妤)

Undergraduate Researcher · Machine Learning · Few-Shot Learning

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SUMMARY

Undergraduate machine learning researcher with hands-on experience in few-shot learning and UAV remote sensing at National Dong Hwa University. Co-developed and published a CAML-based pipeline for forest carbon sink estimation at ACM BDSIC 2025, earning Best Presentation Award. Seeking a **machine learning or ML scientist internship** to apply research expertise in applied ML settings across industry in Asia. Also interested in AI safety and alignment as an emerging research direction.

SKILLS

ML & Deep Learning	Few-Shot Learning, CNN, ResNet50, EfficientNet, GANs, PyTorch, Episodic Training, Scikit-learn, XGBoost, LightGBM
Remote Sensing & GIS	QGIS, GeoTIFF, Raster/CHM Processing, UAV RGB Imagery, Rasterio
Computer Vision	Image Processing, Image Classification, Feature Extraction, Segmentation
Programming	Python, C, C++, SQL
Data & Visualization	Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Google Colab, SHAP, OpenCV
Dev Tools	Git, GitHub, VSCode, Linux CLI
Cybersecurity	Network Security, SIEM Tools, IDS, Threat Analysis, Linux, Security Automation (Python)

RESEARCH EXPERIENCE

Undergraduate Research Assistant — Few-Shot Learning

ICMN Lab, National Dong Hwa University

Sep 2025 – May 2026

Carbon Sink Estimation in Primary Forest via Tree Segmentation and Species-Level Classification

- Built and implemented a **custom CAML few-shot learning pipeline** (ResNet50 backbone) for UAV-based tree species classification at **Danongdafu Forest Park**, achieving **70.72% accuracy** under **severely limited labeled data conditions**
- Achieved **92.59% carbon sink estimation accuracy** vs. ground-truth survey — supporting Taiwan’s 2025 carbon fee policy as a **scalable, cost-effective alternative** to traditional field surveys costing **NTD 5M+/30 years**
- Preprocessed GeoTIFF UAV imagery, performed crown annotation, and validated geospatial data using **QGIS** and **CHM analysis** for a primary forest dataset in Taiwan
- Structured episodic training experiments; evaluated models on accuracy, precision, recall, and F1-score; contributed to publication accepted at **ACM BDSIC 2025**

Advisor: Prof. Wei-Che Chien · Team member: Aaron Kuo

PUBLICATIONS & PROJECTS

Carbon Sink Estimation via Tree Segmentation & Species Classification — ACM

Oct 2025

Proceedings of the 7th International Conference on Big-data Service and Intelligent Computation (BDSIC 2025), Bangkok, Thailand — Best Presentation Award and Publication
NDHU CSIE Undergraduate Project Exhibition 2025 — Honorable Mention

May 2025

Alzheimer's Disease Prediction Using Machine Learning

Dec 2025

Final project · Big Data Analytics — NDHU

- Evaluated 11 ML classifiers (Random Forest, XGBoost, LightGBM, MLP, SVM, etc.) on structured clinical data; best model (Random Forest) achieved **96.74% test accuracy** and **95.39% F1-score** on the minority Alzheimer class with **5-fold CV**
- Developed EDA-driven feature selection and engineering pipeline combining **Chi-square tests, t-tests, and Pearson/Spearman** correlation analysis with **heatmap visualization** across 35 clinical features; reduced to 5 discriminative features while preserving clinical interpretability
- Addressed class imbalance (1,389 vs 760 samples) via **scale_pos_weight** and stratified split; analyzed feature importance using Gini, Normalized, and Cumulative perspectives; extended with **SHAP** for model interpretability

Advisor: Prof. Chung Yung · TA: Demiah Kisla Charlery · GitHub available on request

EDUCATION

B.S. Computer Science and Information Engineering (International Program) Sep 2022 – Jul 2026 (Expected)

National Dong Hwa University, Taiwan

GPA: 3.12 · Relevant coursework: Introductory Machine Learning, Big Data Analytics, Operating Systems, Introduction to Generative AI, Project Management

CERTIFICATIONS

Google Cybersecurity Professional Certificate — *Coursera*

Jul – Dec 2024

Google X NTUT (國立臺北科技大學) Learning Program · 8 cybersecurity certificates covering network security, threat analysis, SIEM tools, IDS, and security automation with Python · Verify: coursera.org/verify/professional-cert/5PODXW5LD8CW

WORK EXPERIENCE

Tech Solution Consultant / SEO Content Creator (Intern)

May – Jul 2022

Devfinite Solutions Co., Ltd., Bangkok

- Produced SEO-optimised technical content that improved organic search rankings for client websites; conducted keyword research and trend analysis using industry tools
- Translated complex technical concepts into clear client-facing documentation; managed content pipelines in WordPress CMS and collaborated directly with clients on deliverables
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LANGUAGES

Thai & Lao: Native · **English:** Professional (TOEIC 825) · **Chinese (Traditional):** Basic · **Spanish:** Basic